SE 216 – SOFTWARE PROJECT MANAGEMENT
Spring 2025-2026
Project Proposal

RePlate

Problem Definition

Food waste is a growing problem in urban environments. Restaurants, cafes, bakeries, and supermarkets frequently discard significant amounts of food that remain safe for consumption. This situation usually occurs due to overproduction or the inability to sell certain products before the end of the day. As a result, large quantities of edible food are wasted. At the same time, many individuals with limited financial resources struggle to access affordable meals. This creates a clear imbalance between the availability of surplus food and the people who could benefit from it. Municipalities and local authorities attempt to address food waste through various social aid programs. However, these initiatives often lack an efficient digital infrastructure that allows food businesses, local authorities, and citizens to coordinate effectively. Most existing food delivery platforms focus on selling freshly prepared meals and do not provide a structured system for redistributing surplus food. Consequently, businesses lose potential value while edible food continues to be discarded. Therefore, there is a need for a digital platform that enables the efficient reporting, monitoring, and redistribution of surplus food within local communities.

Background Information

RePlate is a digital platform designed to reduce food waste while also supporting social aid initiatives within the community. The platform aims to create a coordinated food redistribution network by connecting restaurants, food businesses, municipalities, and citizens through a single application. Through the RePlate platform, food businesses such as restaurants, cafes, bakeries, and grocery stores will be able to report their surplus food instead of discarding it. Businesses will enter information such as food type, quantity, availability duration, and pickup location through the application. The system will operate through a centralized application structure where all food reports will be processed and stored. Available food listings will be presented to users through an interactive map interface that displays nearby restaurants and available food items.

The platform will operate using a two-stage redistribution model. In the first stage, businesses will report surplus food through the application and the food will remain available at the restaurant for a limited period of time. During this period, users will be able to view the available

products through the application, reserve suitable items, and collect them directly from the restaurant. If the food is not collected within the specified time, the system will move the listing to the second stage of the redistribution process. In this stage, the remaining food will be transferred to designated municipal distribution centers. Users will be able to view nearby centers and available food through the application and collect products from these locations. Municipalities will manage this process through an administrative panel where they can monitor reported food, track transfers to distribution centers, and manage redistribution operations. The platform will also support donation programs. Users will be able to donate meals for people in need through the application, and these meals will be delivered through municipal social aid centers. In addition, users will be able to donate pet food that can be delivered to animal shelters. In addition to the redistribution process, the system will provide analytical tools for businesses and municipalities. Businesses will be able to monitor surplus food reports and waste reduction statistics through a dashboard. Municipalities will be able to analyze regional food redistribution data through a management interface.

In future development phases, an AI-based prediction module may be integrated into the system. This module could estimate which products are likely to remain unsold based on historical data and help businesses reduce overproduction.

Objectives

- Reduce food waste by enabling restaurants and food businesses to report surplus food instead of discarding it
- Create a coordinated digital platform connecting businesses, municipalities, and citizens
- Allow users to locate nearby surplus food through an application interface
- Enable users to reserve and collect food directly from restaurants within a limited time period
- Transfer uncollected food to municipal distribution centers for further redistribution
- Provide a map-based interface that displays nearby food sources and distribution centers
- Support meal donation programs for individuals in need
- Support pet food donation programs for animal shelters
- Provide analytical dashboards for businesses and monitoring tools for municipalities
- Explore AI-based prediction systems to estimate surplus food in future system versions

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Approval Signatures and GitHub Accounts

Group GitHub Account: https://github.com/Se216Section1Team7

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